

Biophysically-inspired modeling of large-scale brain activity

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Overview

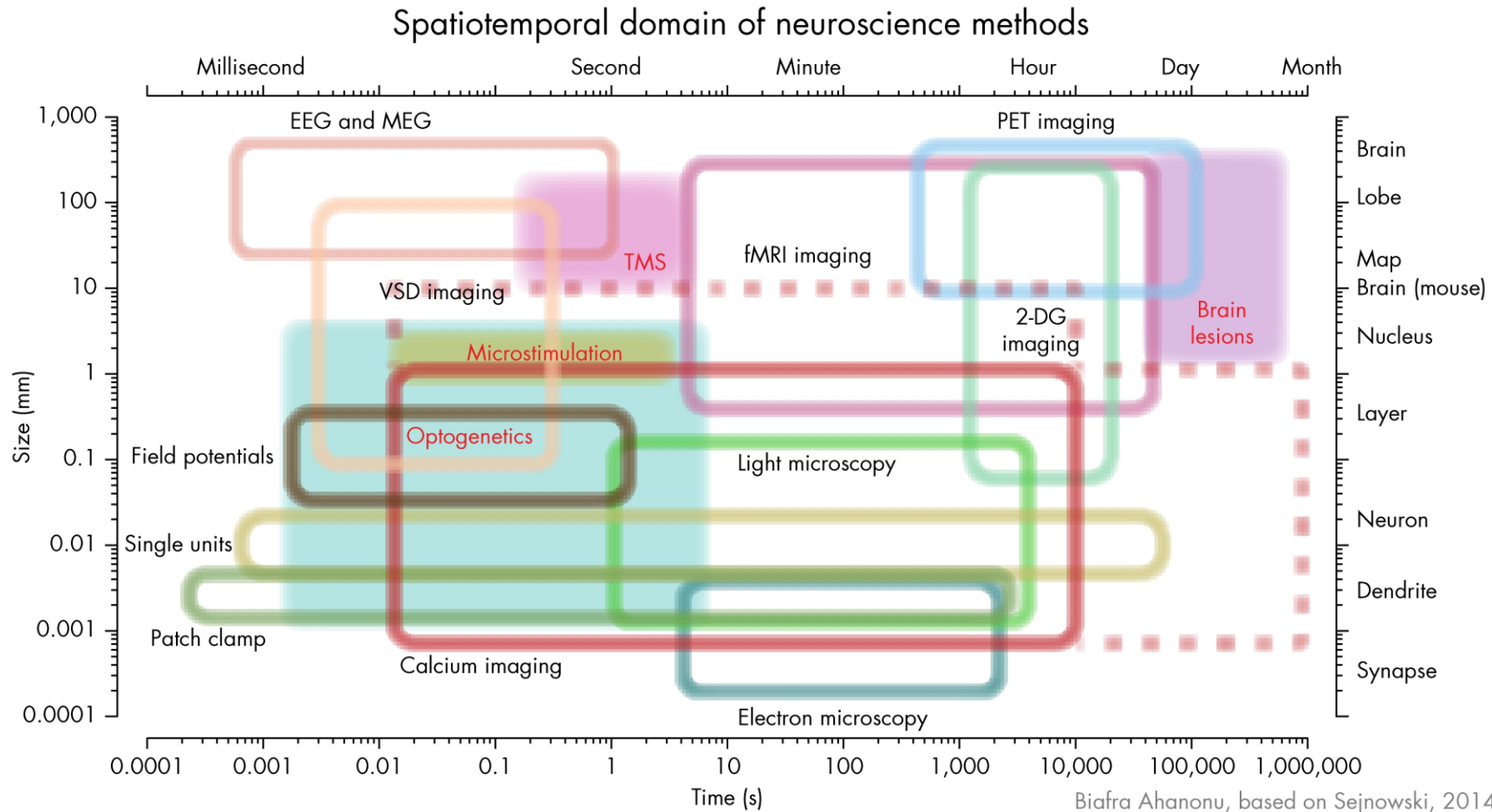
- Applications of large-scale modeling and Whole-Brain modeling
- How to Model: (from neurons to) the brain
 - Dynamics of the brain (...*areas*): reducing dimensions
 - Notable models:
 - Wilson-Cowan
 - Jansen & Rit
 - Wong-Wang MFM.

Why model?

- We study the brain with many different tools

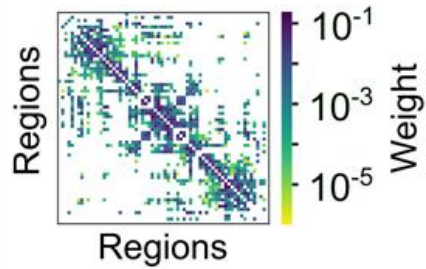
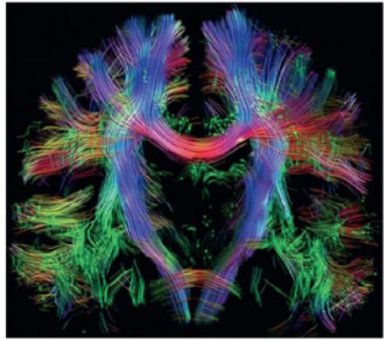


- Molecular / Cellular Biology
- Anatomy
- Imaging – structure
- Imaging – function (fMRI)
- Electrophysiology (EEG)
- Psychology and Behavior

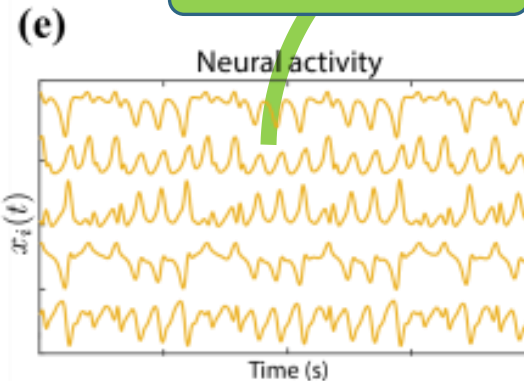
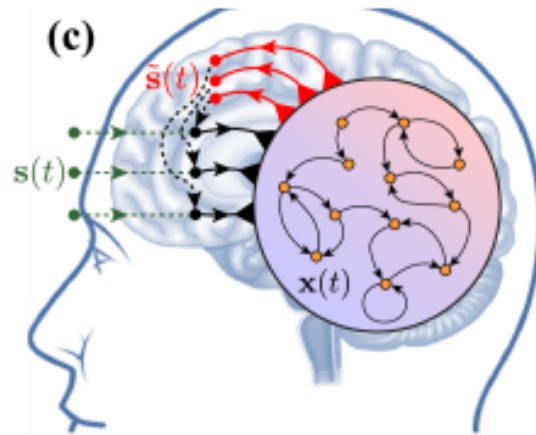


Our brain is a *dynamical* system

Structural Connectivity

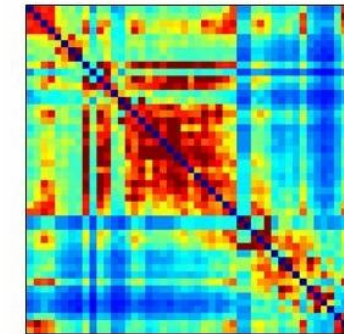
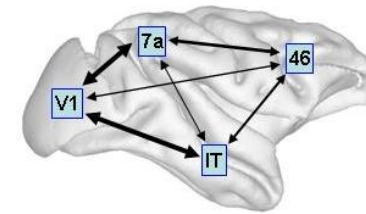


+
Neural dynamics



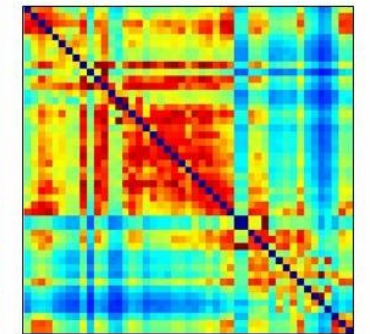
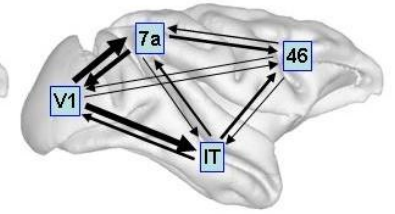
Lu & Basset 2018

Functional Connectivity



Correlation
Mutual Information
Coherence
etc.

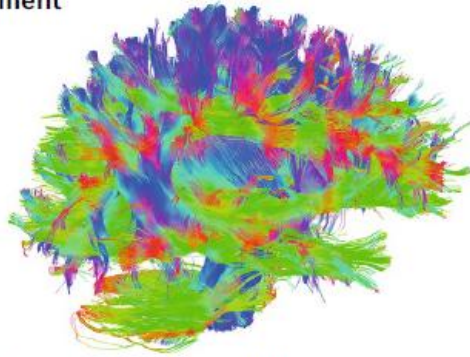
Effective Connectivity



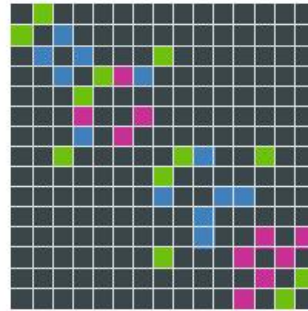
Granger causality
Transfer Entropy
Time-delayed correlation
etc.

Structural vs. Functional

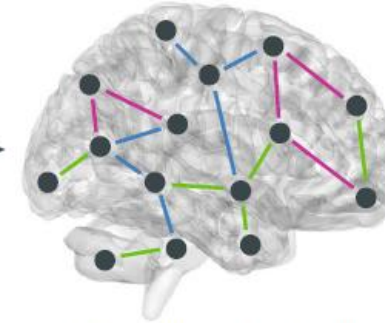
a Measurement



Example: white matter tracts (via diffusion tensor imaging)

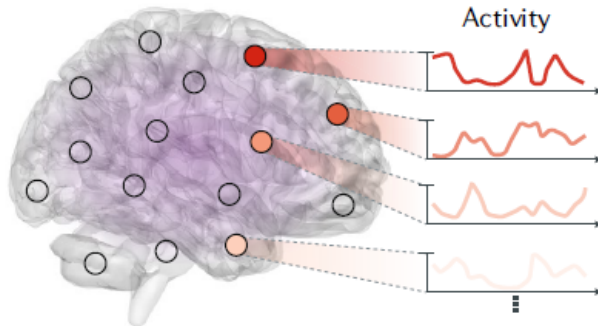


Adjacency matrix

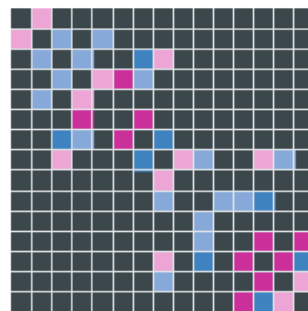


Structural brain network

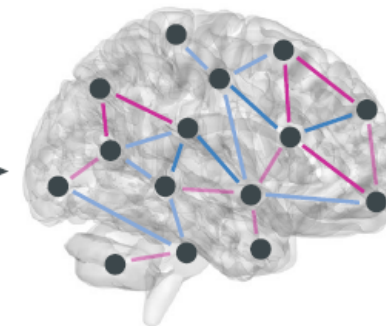
a Measurement



Example: blood oxygen level (via fMRI)

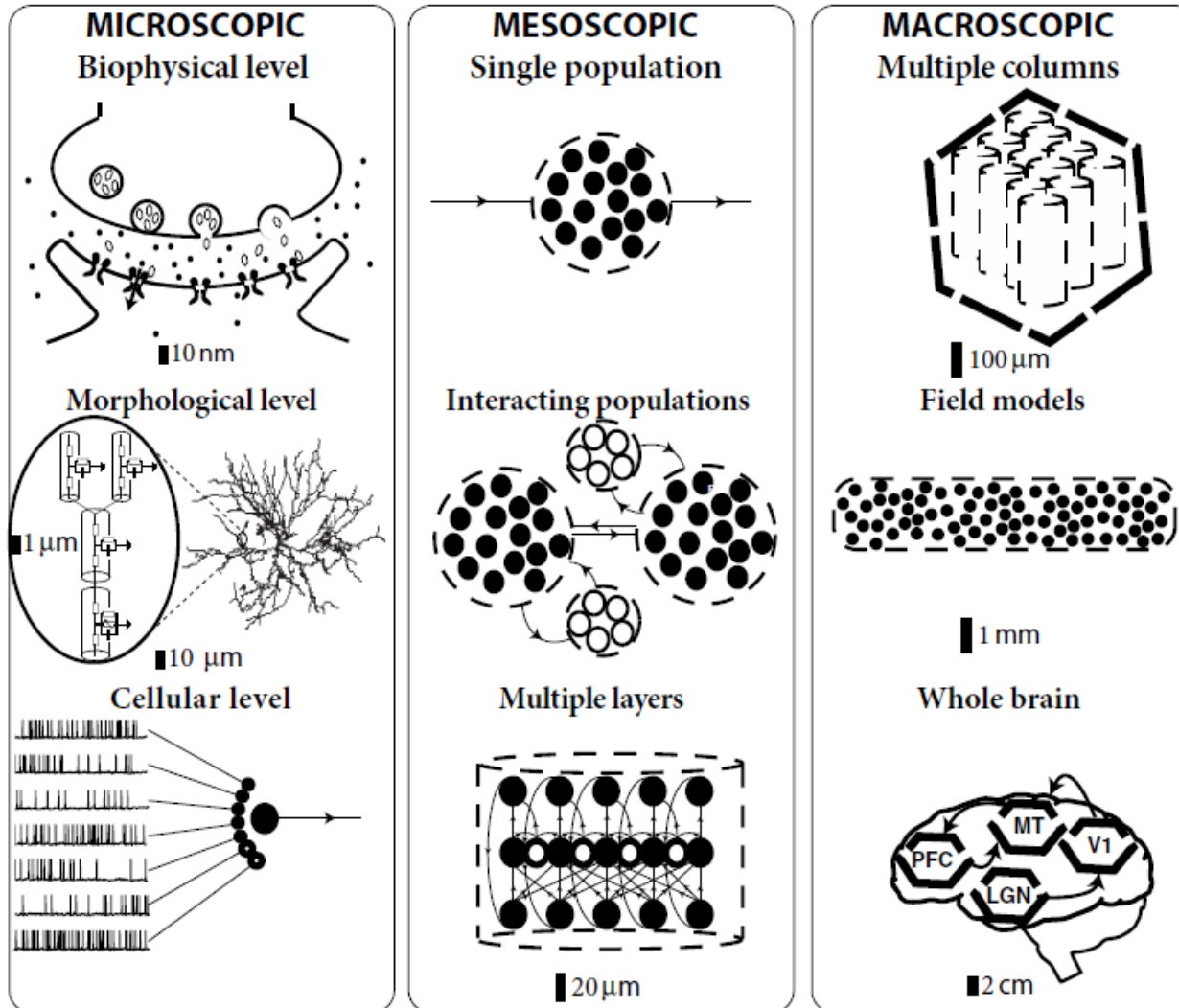


Similarity matrix



Functional brain network

Models at different scales

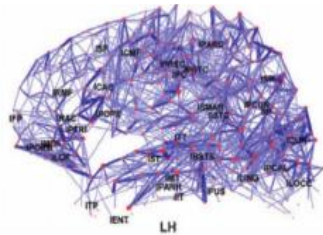


‘Models’ in the physics sense
- A set of rules that predict the time evolution (or the equilibrium statistics) of a system, *based on known or assumed physical laws.*

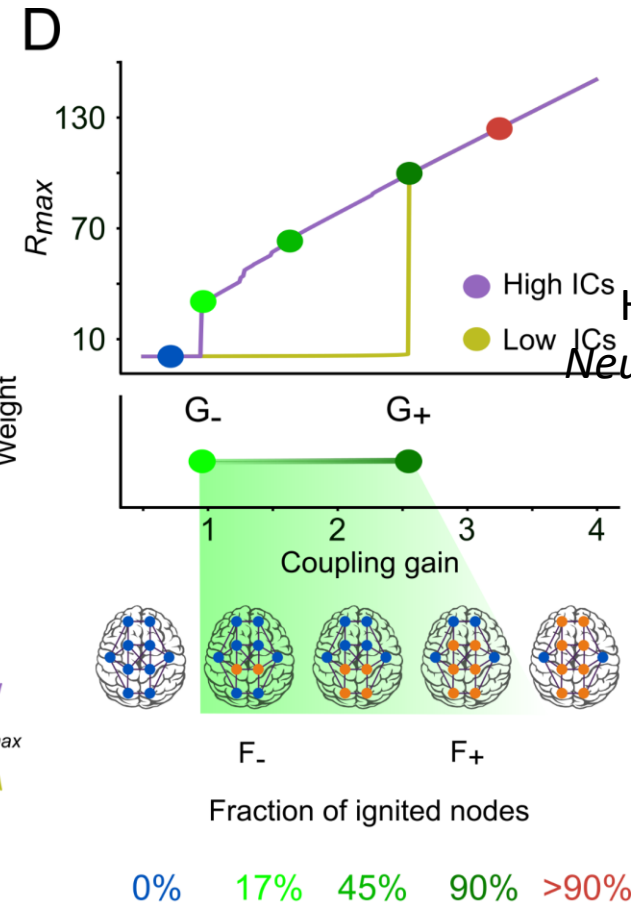
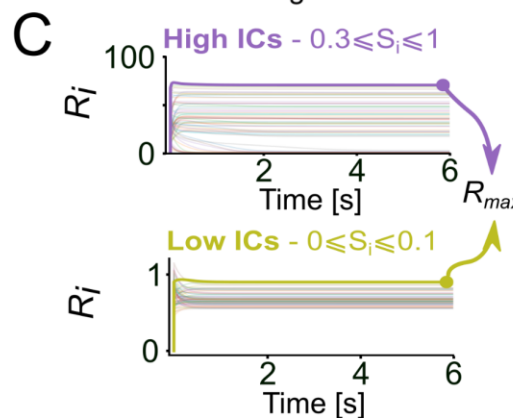
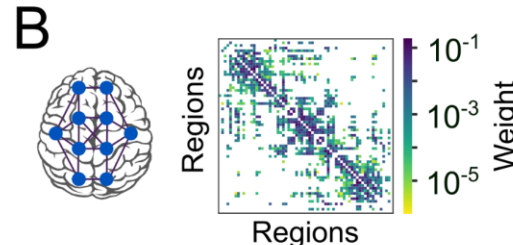
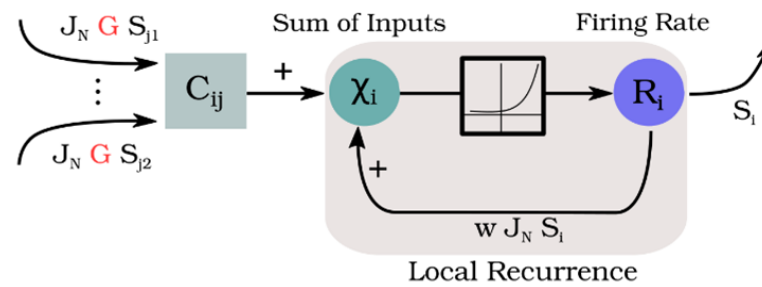
Applications of large-scale modeling

Human connectome + dynamical model

= bistability



Hagmann *et al.*,
PLoS Biol (2008)



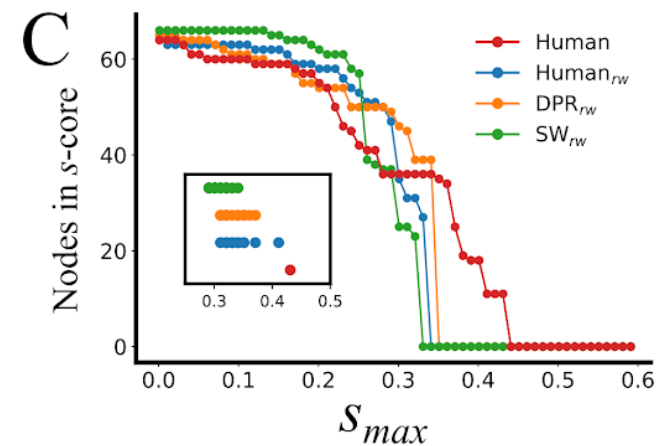
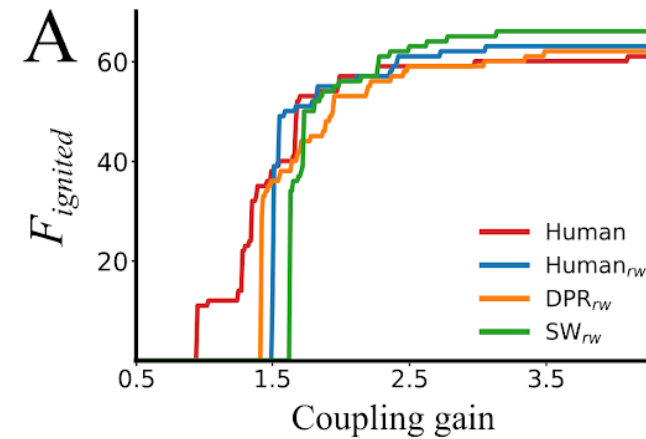
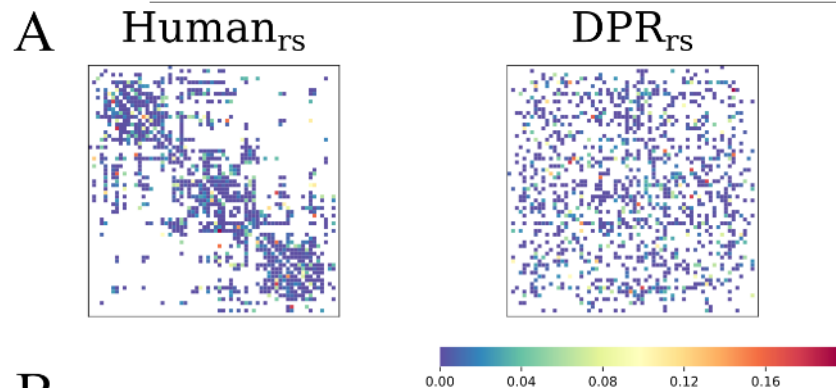
Hansen *et al.*,
Neuroimage (2015)

RESEARCH ARTICLE

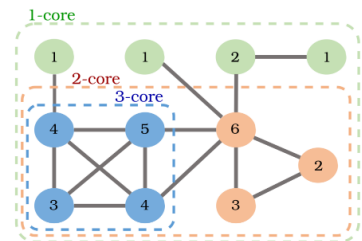
Cortical ignition dynamics is tightly linked to the core organisation of the human connectome

Samy Castro^{1,2†}, Wael El-Deredy³, Demian Battaglia^{4†*}, Patricio Orio^{1,5†*}

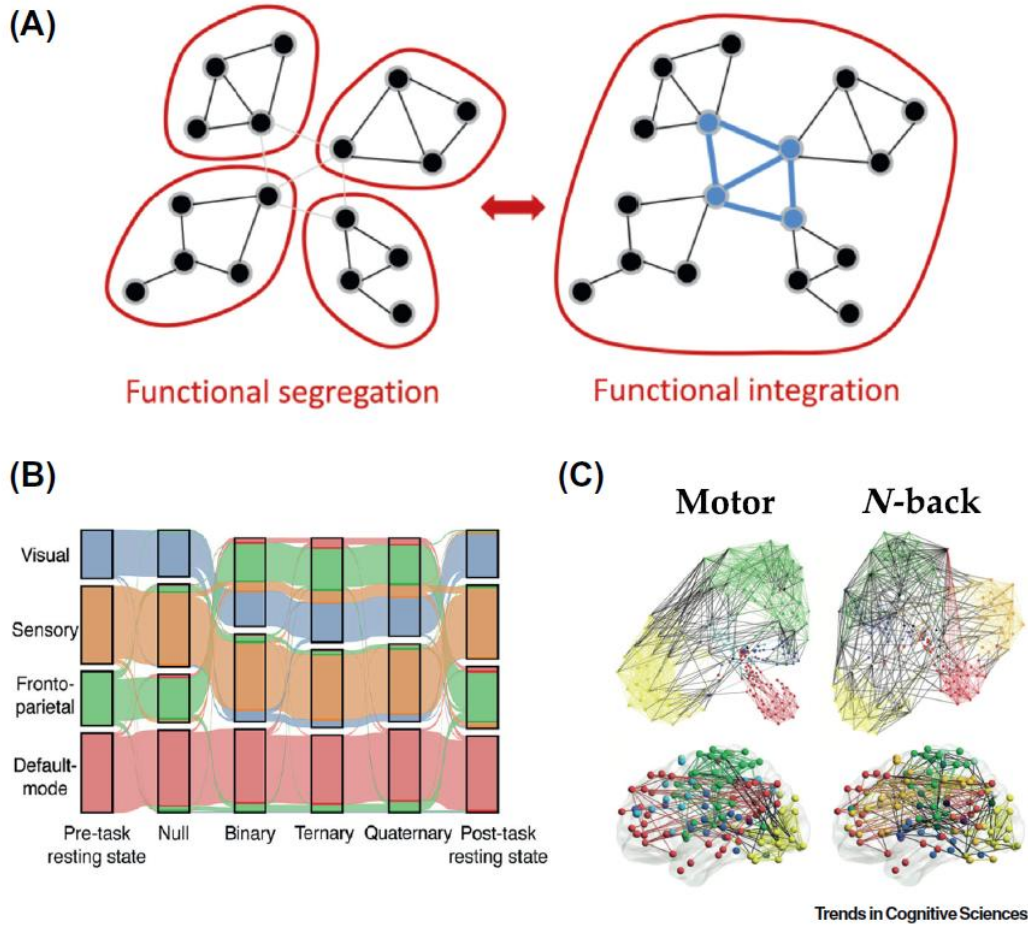
Bistability in random weighed models



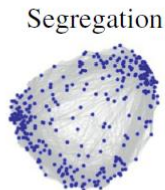
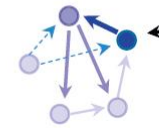
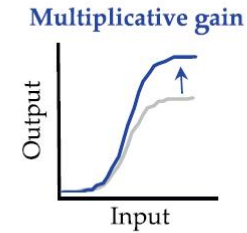
“Ignition core”



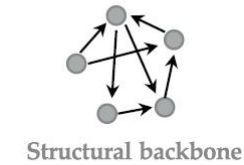
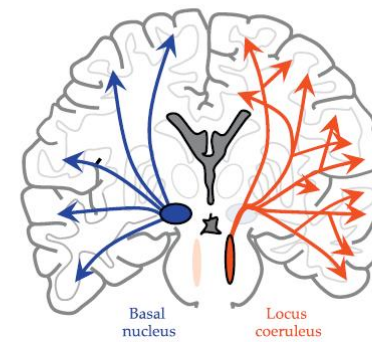
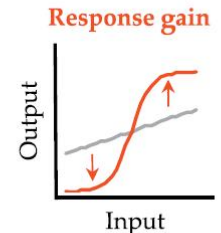
Neuromodulation and the Integration/Segregation balance



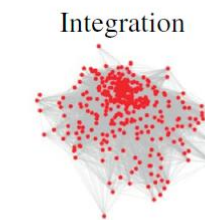
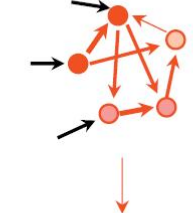
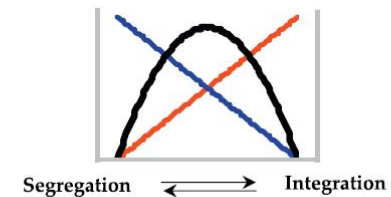
Cholinergic system



Noradrenergic system



Inverted-U relationship

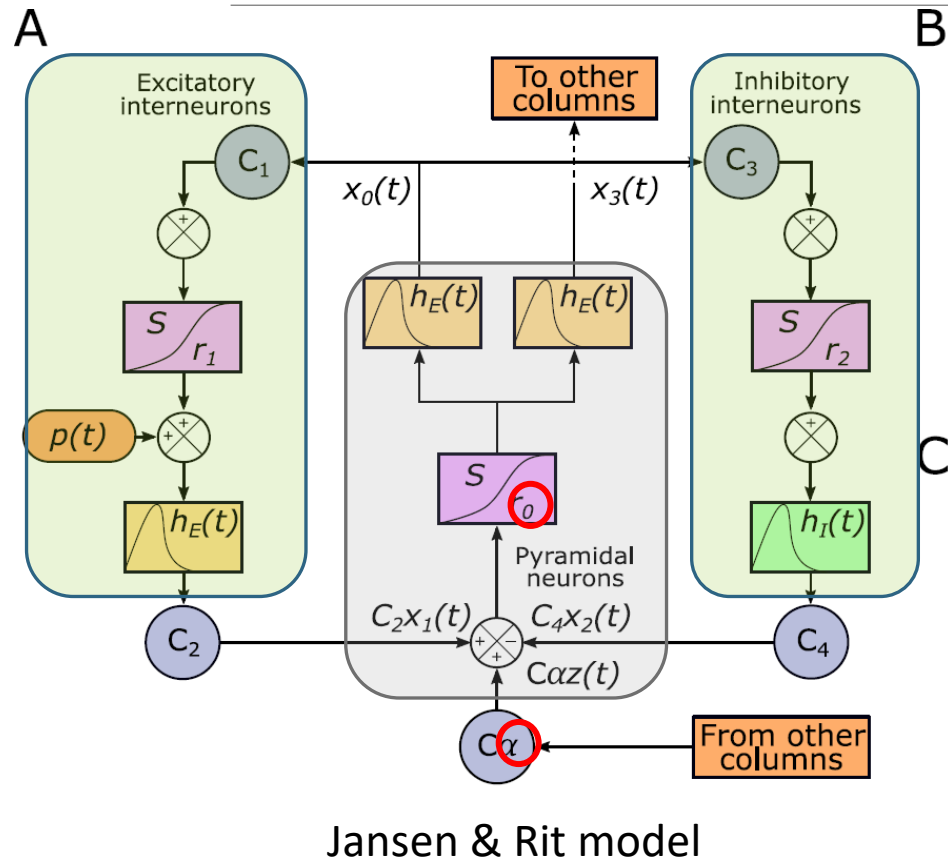


Trends in Cognitive Sciences

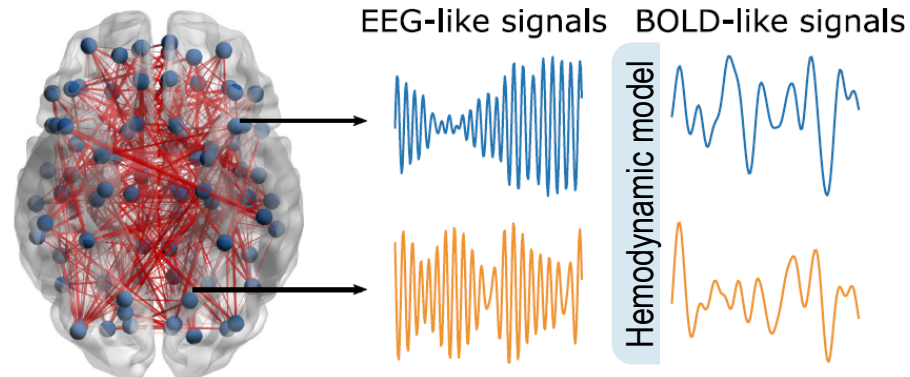
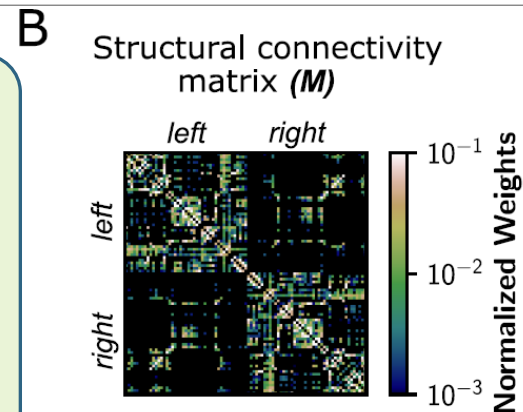
Simulating the Integration/Segregation switch



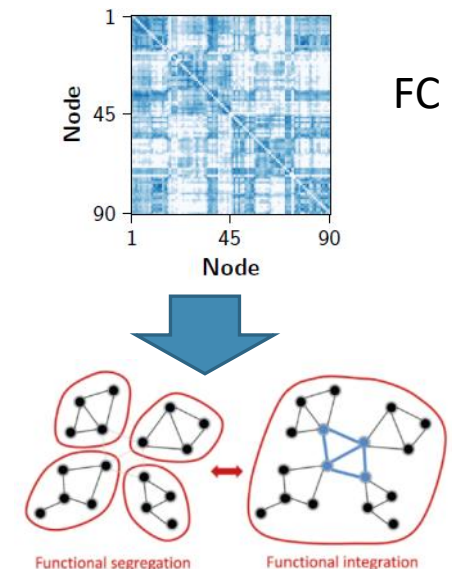
Carlos Coronel



α : Cholinergic system
 r_0 : Noradrenergic system

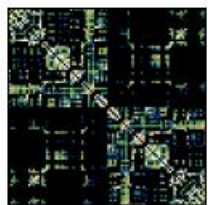


Graph analysis
 Efficiency E^w
 Modularity Q^w

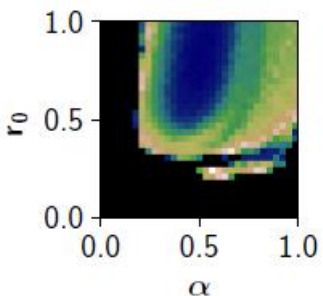
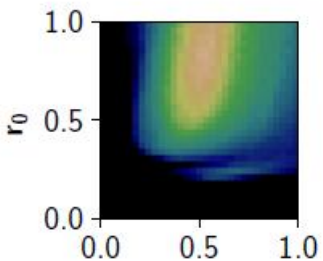
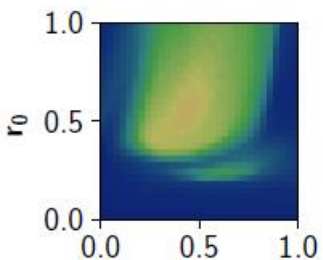


A

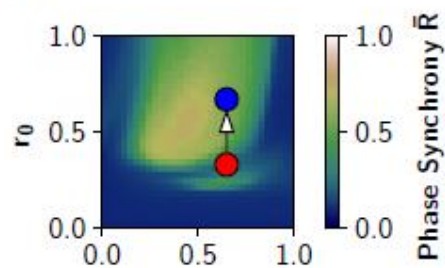
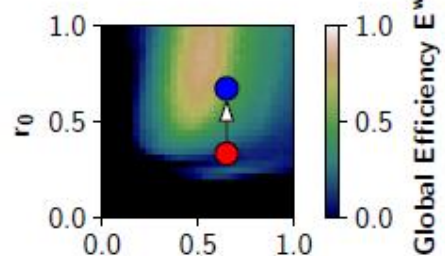
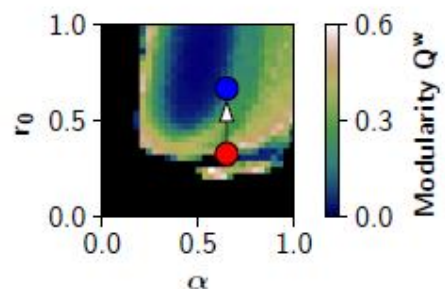
Human



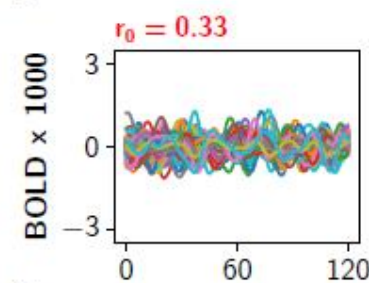
Normalized Weights

Phase Synchrony $\bar{R}$ Global Efficiency E^w Modularity Q^w 

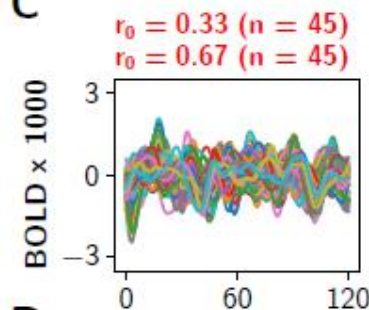
A

Phase Synchrony $\bar{R}$ Global Efficiency E^w Modularity Q^w

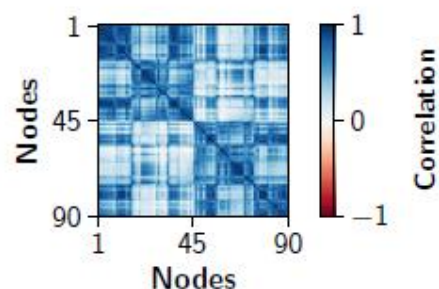
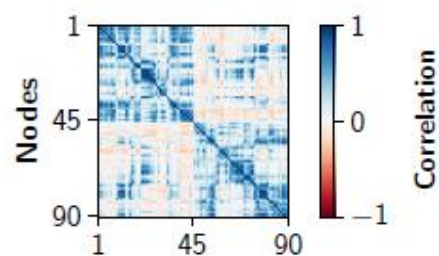
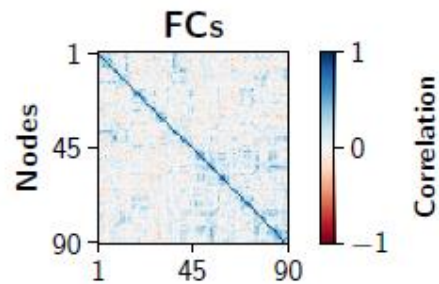
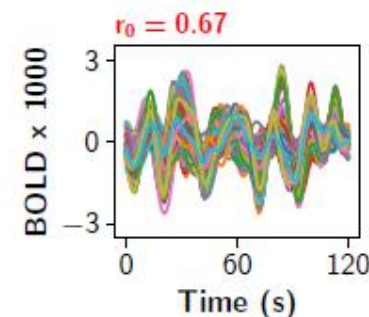
B



C

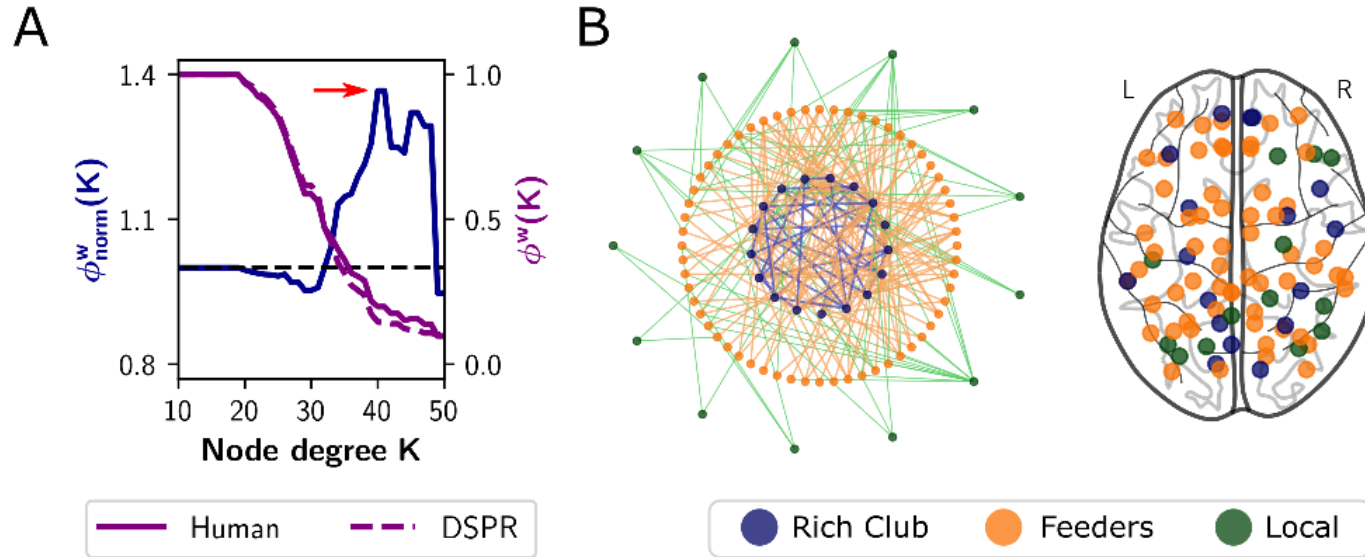


D

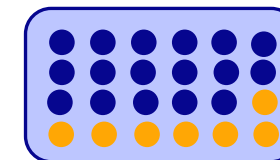


What subset of nodes has the greatest global effect when neuromodulated?

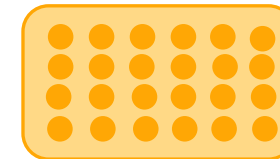
Rich club analysis



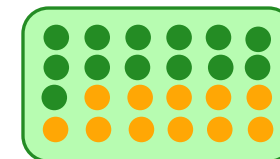
Modulation of rich-club nodes has the highest network impact



17 rich club +
7 random feeders



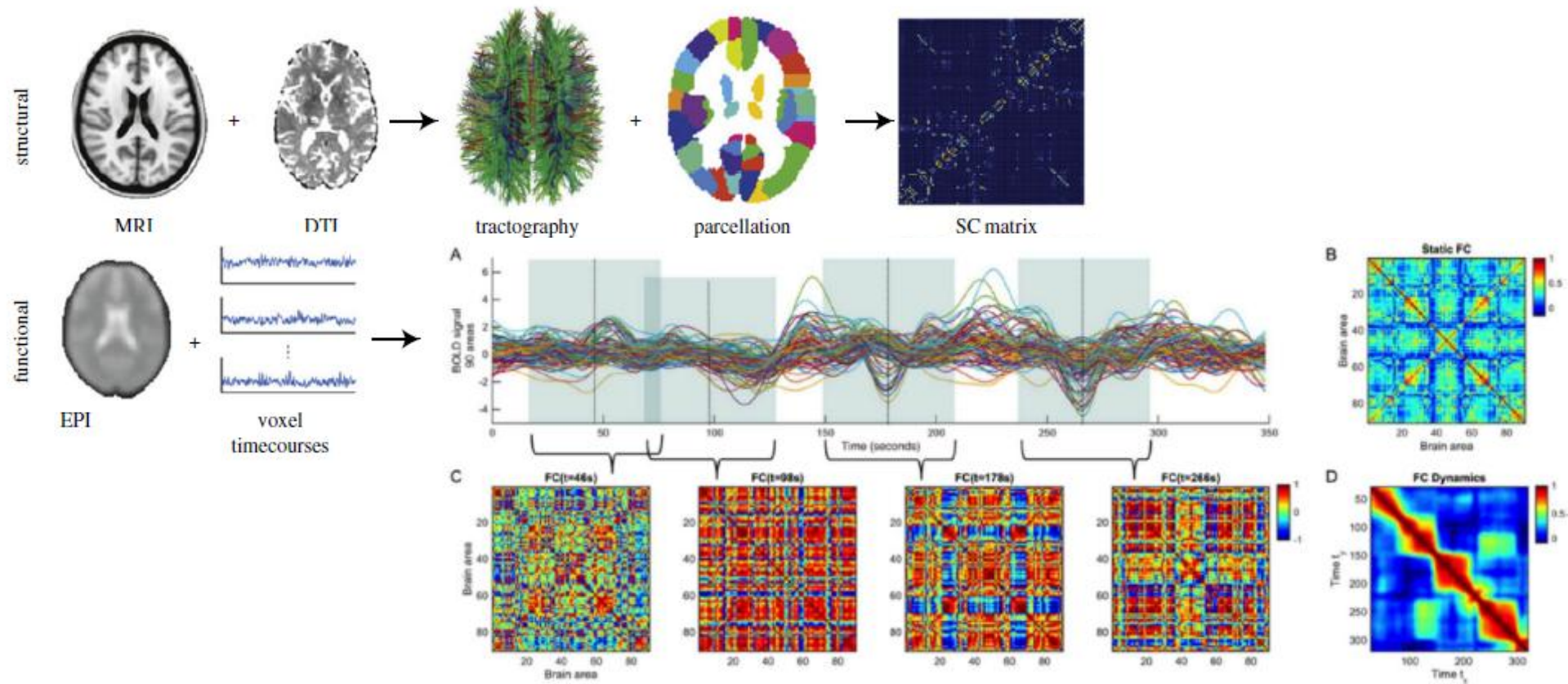
24 random feeders



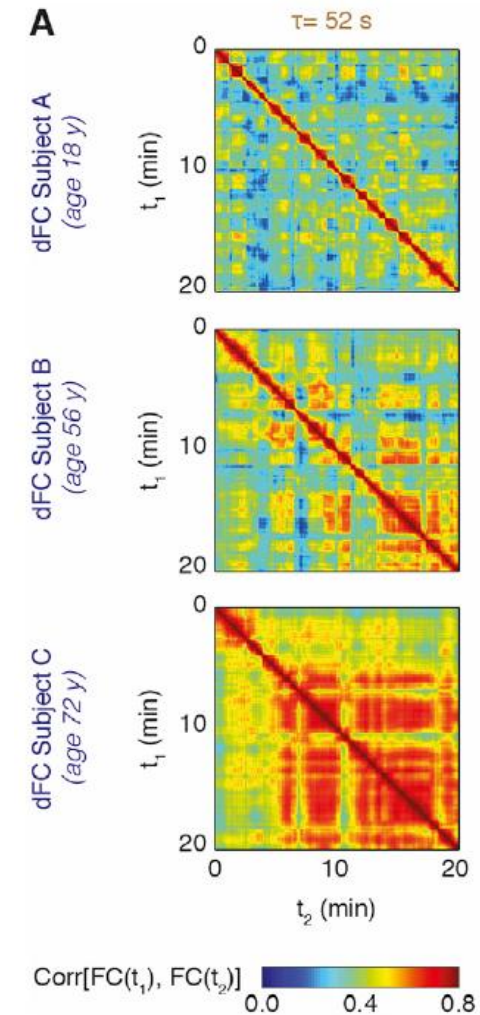
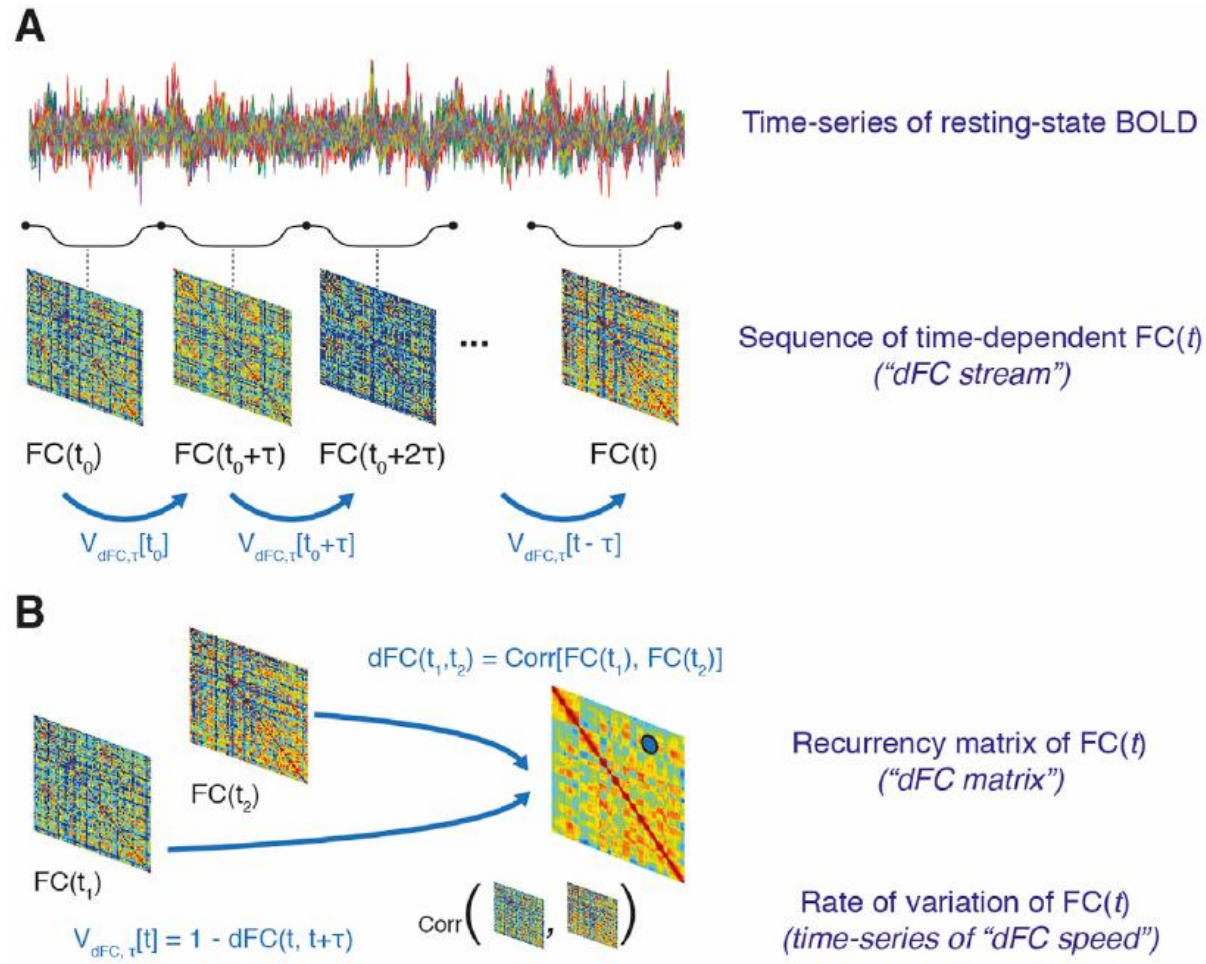
13 local +
11 random feeders

Network topology and FCD

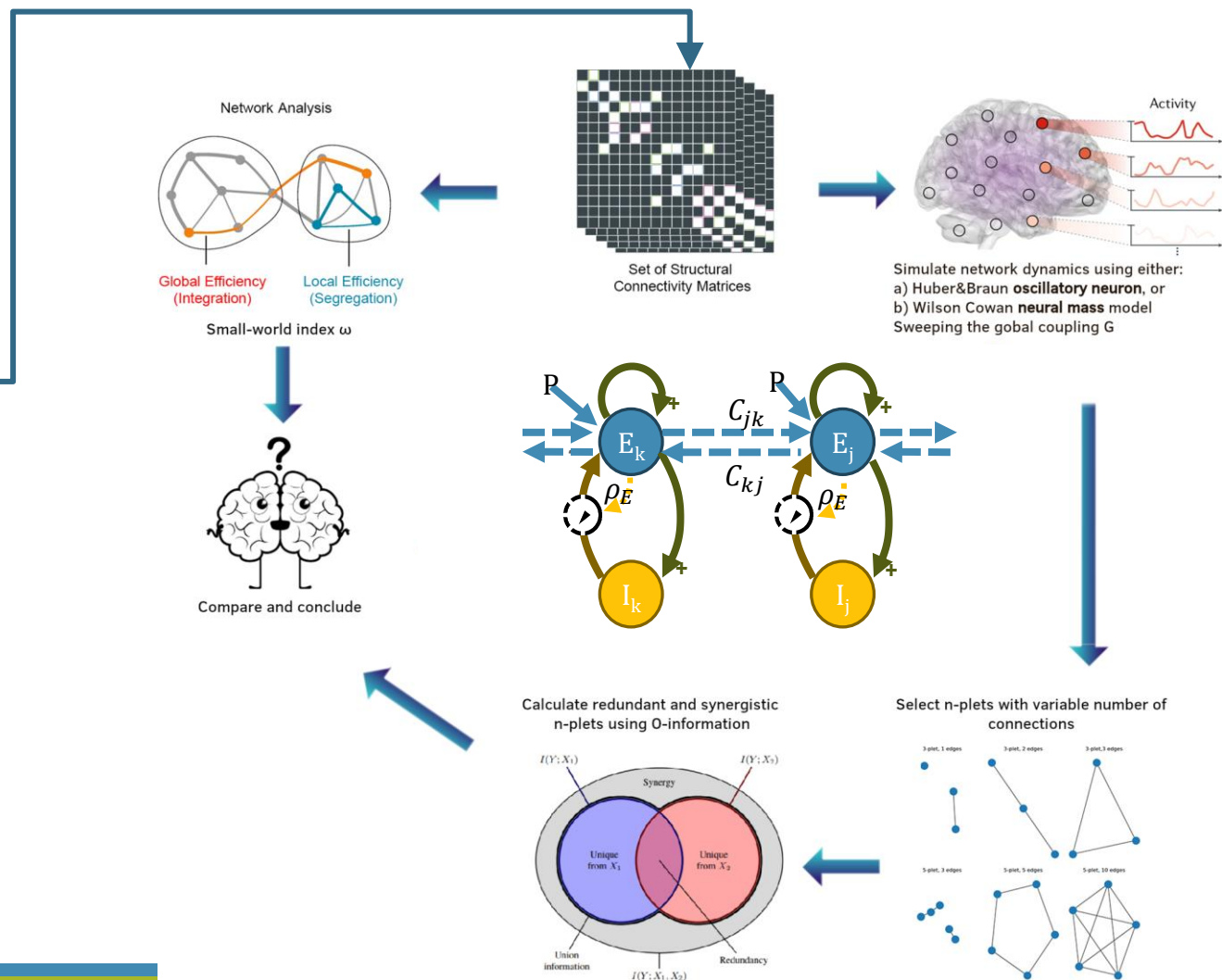
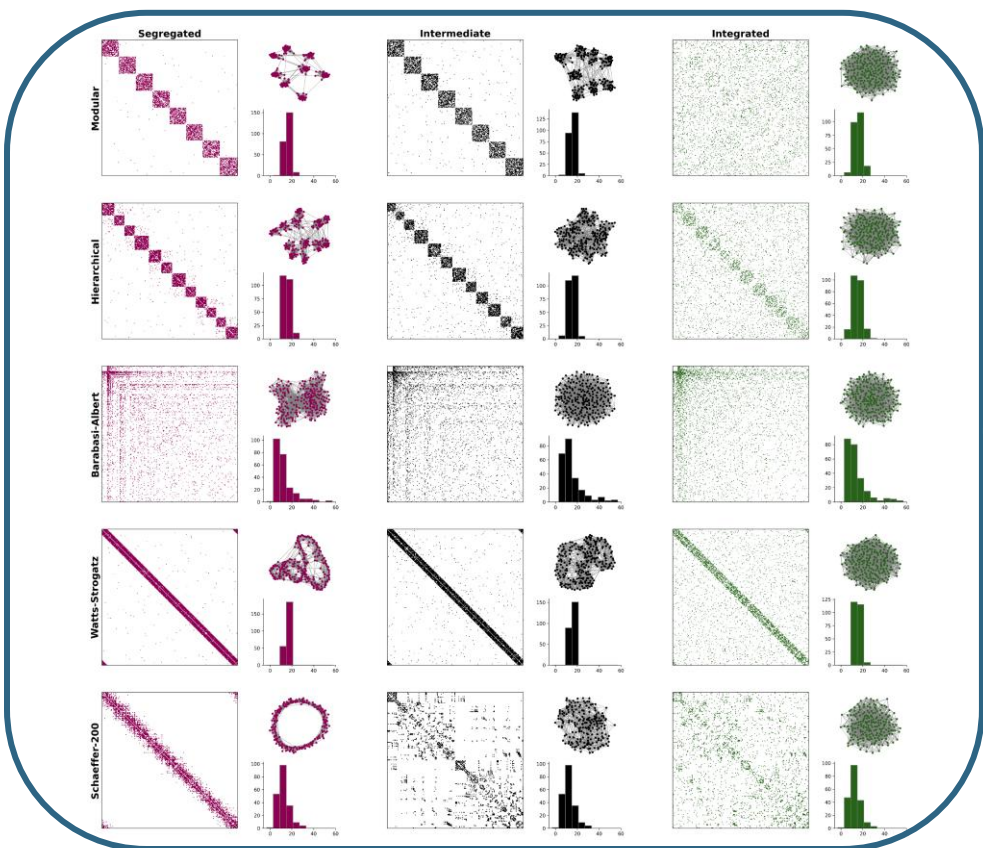
Structural (SC) and Functional (FC) *connectivity*



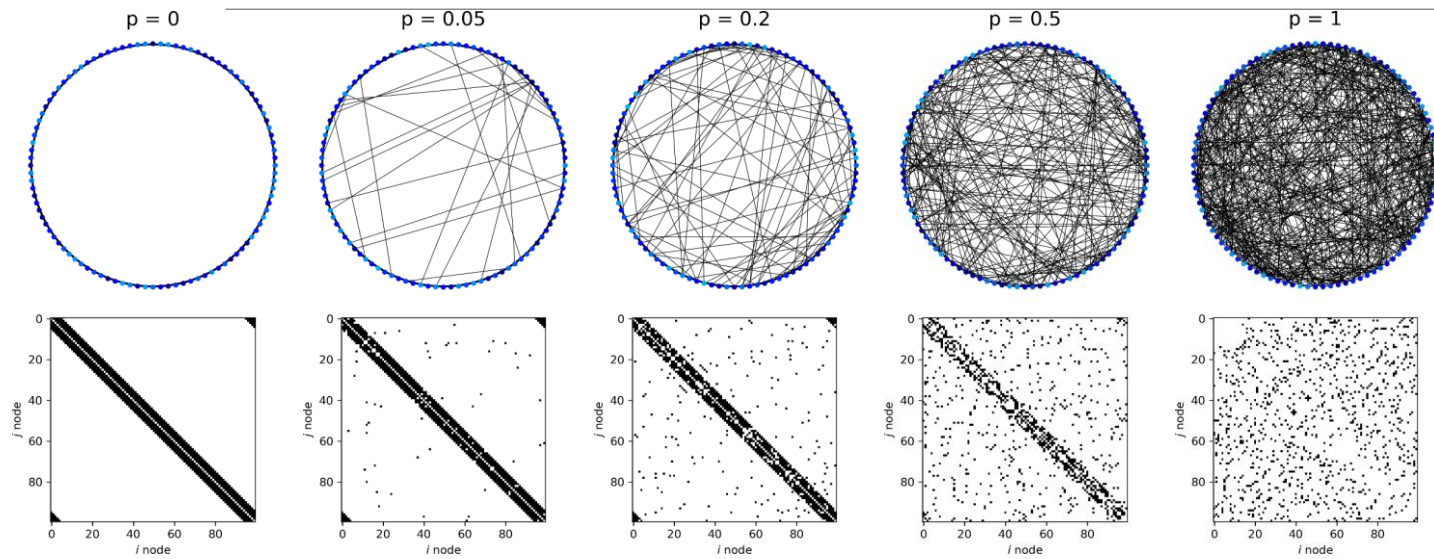
Functional Connectivity Dynamics



Assessing the role of topology



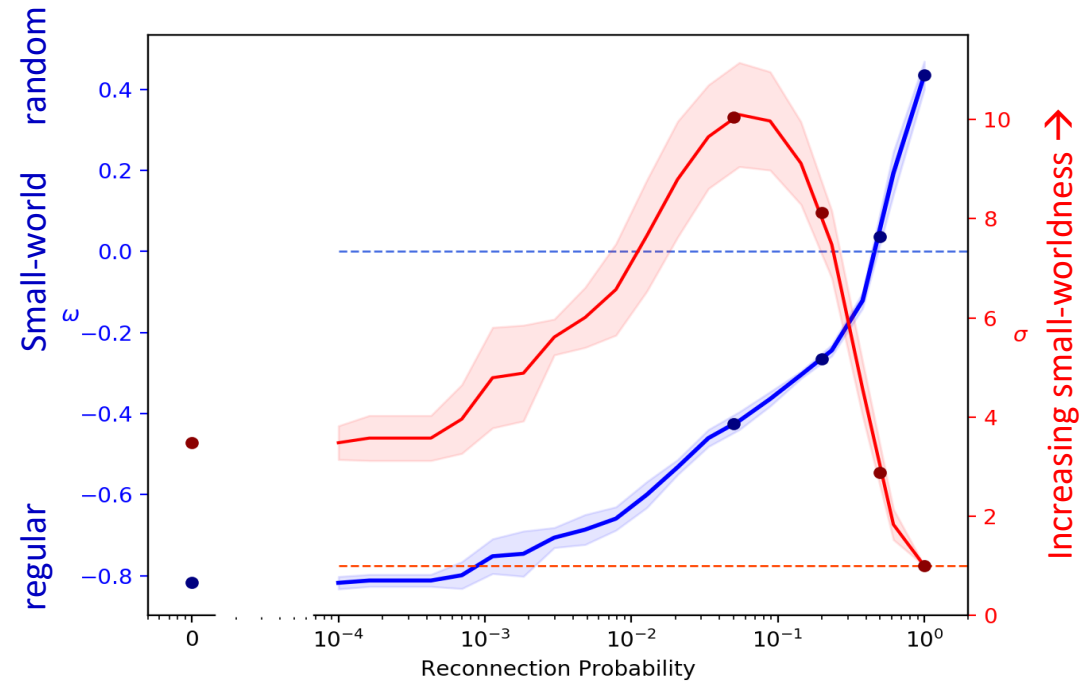
From regular to random

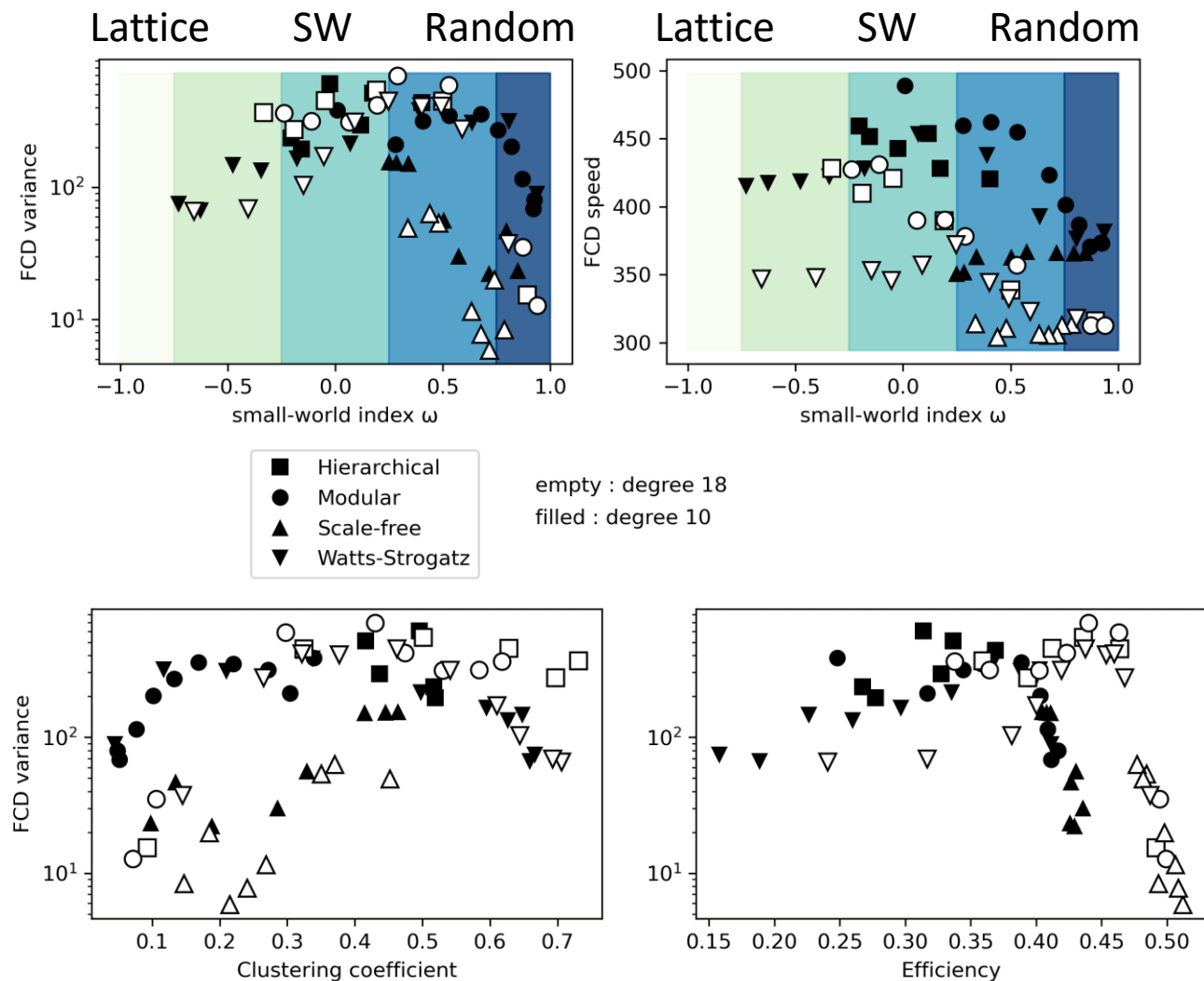
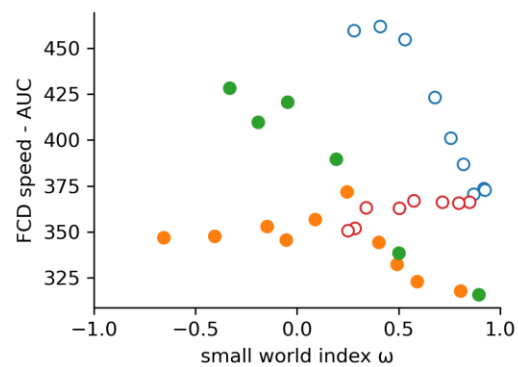
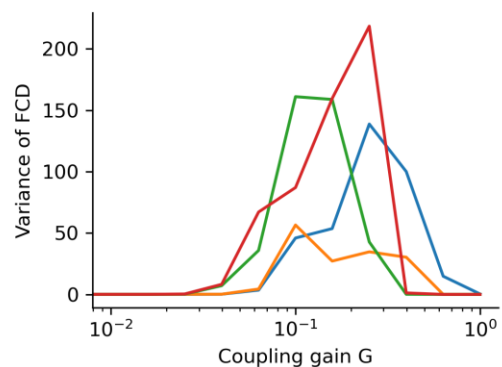
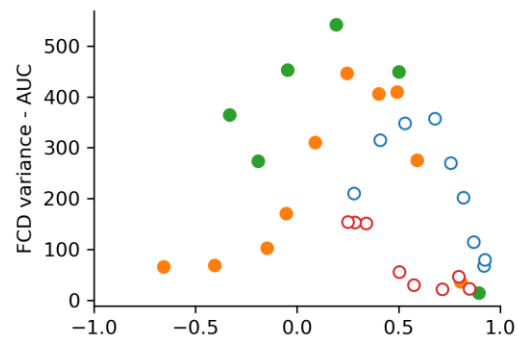
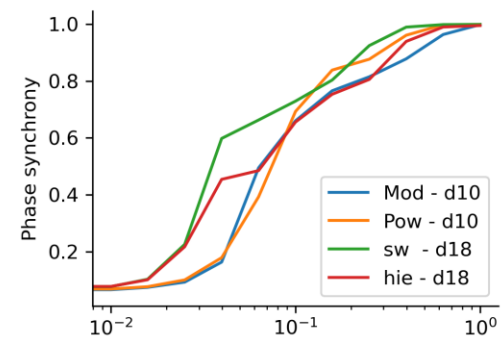


Small world indices:
 σ (Humphries) and ω (Telesford):

$$\sigma = \frac{C/C_r}{L/L_r}$$

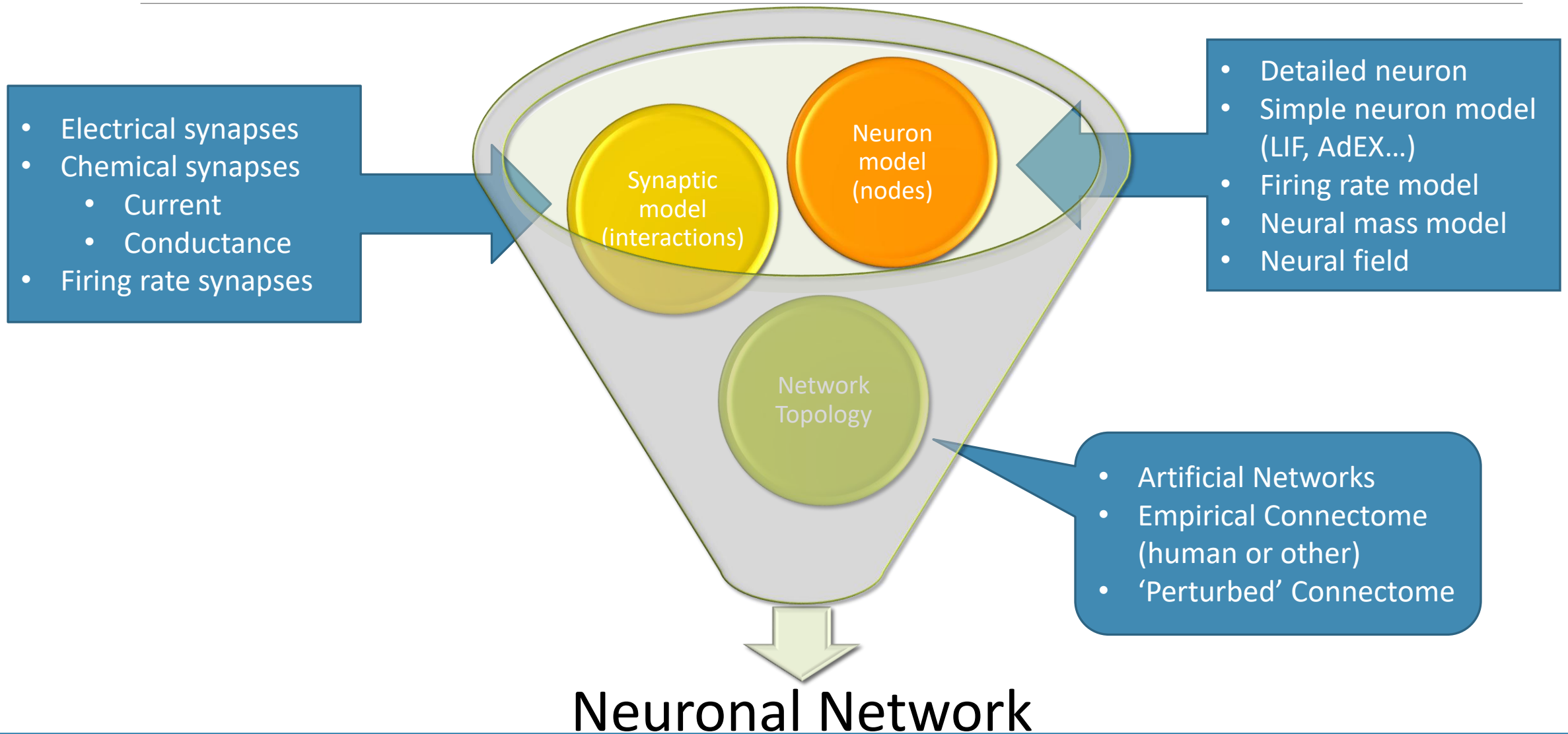
$$\omega = \frac{L_r}{L} - \frac{C}{C_0}$$





Biophysical modeling of brain activity

Network modeling

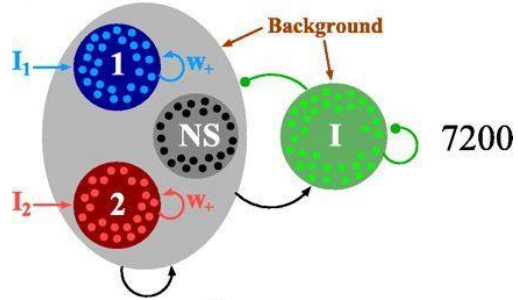


Tools for modelers

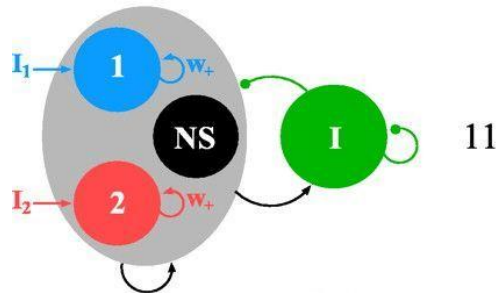
- The Virtual Brain
 - <https://www.thevirtualbrain.org/tvb/zwei/brainsimulator-software>
- Brain Dynamics Toolbox
 - <https://bdtoolbox.org>
- BRIAN Simulator → spiking networks (and more)
 - <https://briansimulator.org/>
- NEURON → Detailed neurons, spiking neurons, networks.
 - www.neuron.yale.edu
- NETPYNE → NETworks in PYthon + NEuron
- DIY in Python or Matlab!!

Mean-field models / Neural mass models

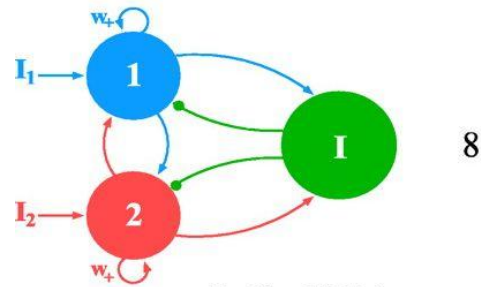
Spiking neuronal network model



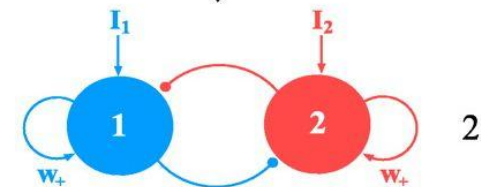
Mean-field approach



Simplified F-I curves
+
Constant activity
of NS cells



Slow NMDA
gating variable



Reduced two-variable model

Wong & Wang, 2006

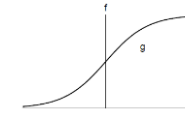
Input from network

External input

Neuron state variable:
Voltage, PSP, gating variable

<pop avg>

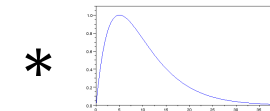
Non-linear function



Firing rate

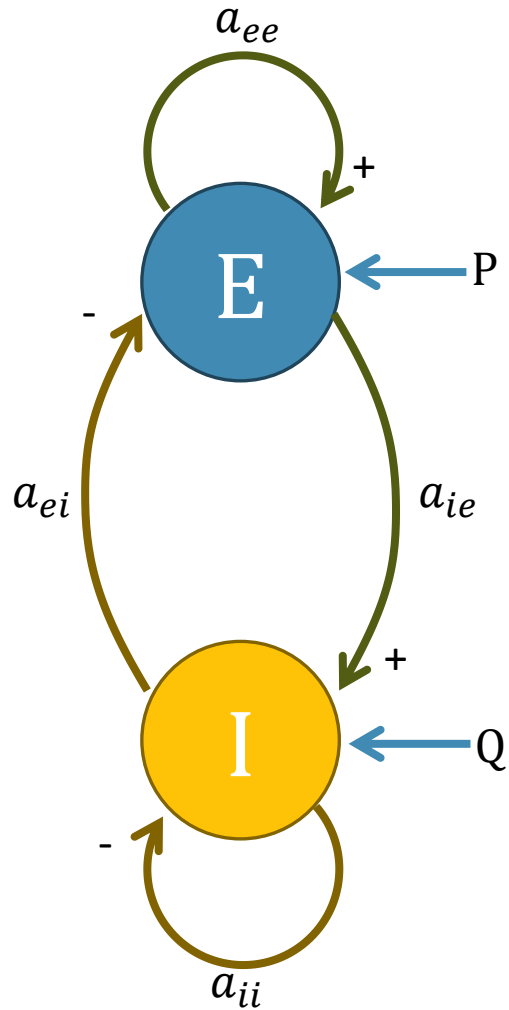
<pop avg>

linear transformation



Output

Wilson-Cowan



- 2 neuron populations: excitatory and inhibitory
- Variables (E,I) represent the firing rate of each population.

$$\tau_e \frac{dE}{dt} = -E + (1 - r_e E) F_e(\alpha_{ee} E - \alpha_{ei} I + P)$$

$$\tau_i \frac{dI}{dt} = -I + (1 - r_i I) F_i(\alpha_{ie} E - \alpha_{ii} I + Q)$$

$$F(x) = \frac{1}{1 + \exp(-\beta(x - x_T))}$$

Analysis of chaotic oscillations induced in two coupled Wilson–Cowan models

Yuya Maruyama · Yuta Kakimoto · Osamu Araki

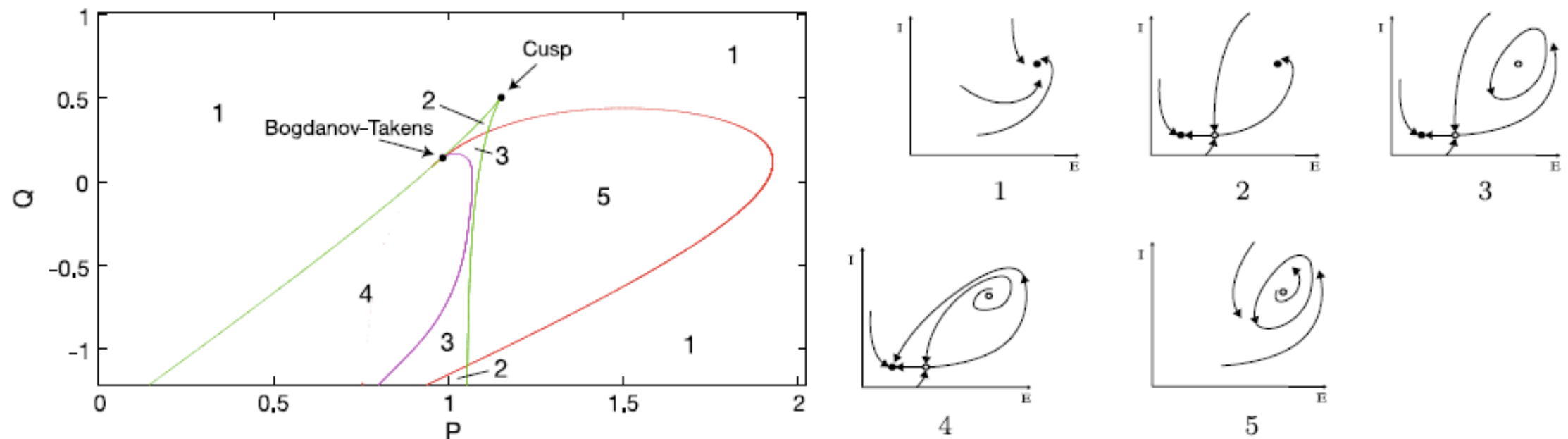
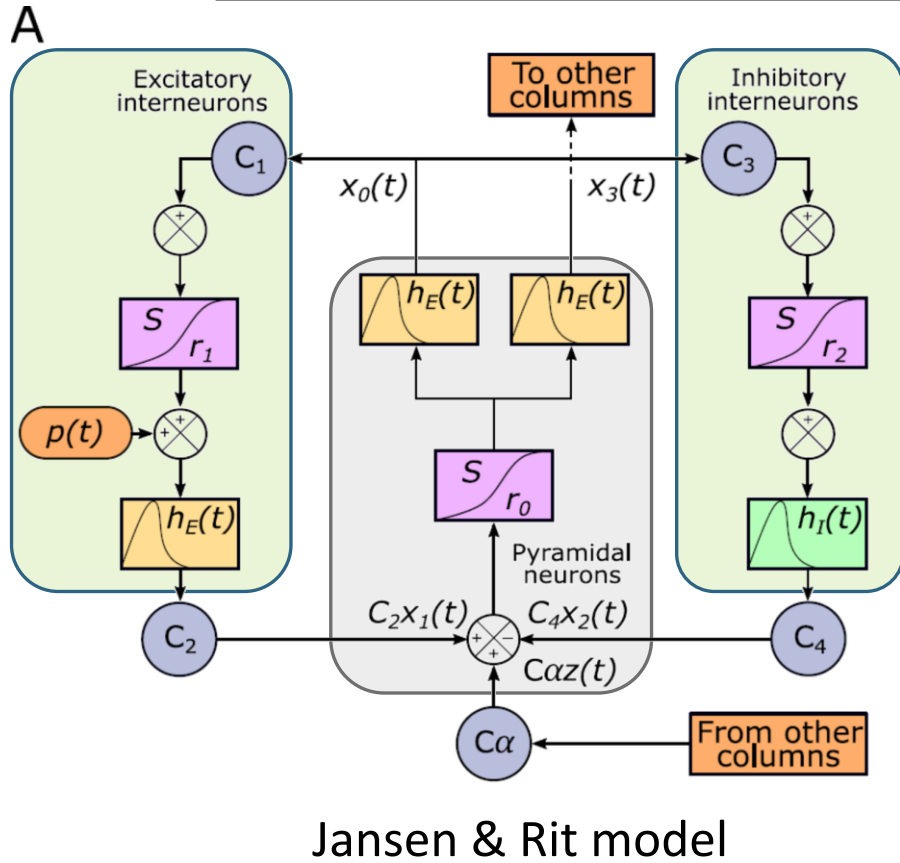


Fig. 2 Bifurcation sets of the Wilson–Cowan oscillator plotted on the P – Q plane (*left panel*) and the schematic phase planes (*right panels 1–5*). Each scheme of phase plane corresponds to the state identified by each number on the P – Q plane. Three types of bifurcation sets, i.e.,

Andronov–Hopf (*red*), saddle-node (*green*), and saddle separatrix loop (*purple*) are plotted. Cusp and Bogdanov–Takens bifurcation points are also shown in the diagram

Jansen & Rit – Neural mass model

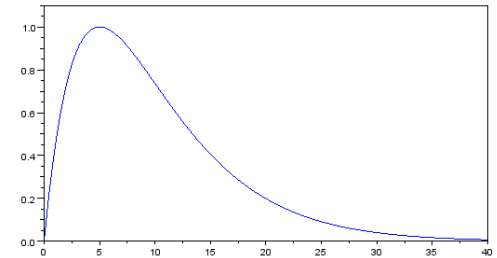


$$x_0'' = AaS(C_2x_1 - C_4x_2) - 2ax_0' - a^2x_0$$

$$x_1'' = Aa(p(t) + S(C_1x_0)) - 2ax_1' - a^2x_1$$

$$x_2'' = BbS(C_3x_0) - 2bx_2' - b^2x_2$$

$$S(x) = \frac{2e_0}{1 + \exp(-r_0(x - x_0))}$$



$$x_0' = x_3$$

$$x_1' = x_4$$

$$x_2' = x_5$$

$$x_3' = AaS(C_2x_1 - C_4x_2) - 2ax_3 - a^2x_0$$

$$x_4' = Aa(p(t) + S(C_1x_0)) - 2ax_4 - a^2x_1$$

$$x_5' = BbS(C_3x_0) - 2bx_5 - b^2x_2$$

Dynamical Mean-field model

(Wong-Wang-Deco; Deco et al. 2013)

$$\frac{dS_i}{dt} = -\frac{S_i}{\tau_s} + (1 - S_i)\gamma R_i + \sigma\eta_i(t)$$

$$R_i = \frac{ax_i - b}{1 - \exp(-d(ax_i - b))}$$

$$x_i = wJ_N S_i + J_N G \sum_j C_{ij} S_j + I_0$$

$$\frac{dS_i^E}{dt} = -\frac{S_i^E}{\tau_{NMDA}} + (1 - S_i^E)\gamma r_i^E + \sigma\eta_i(t)$$

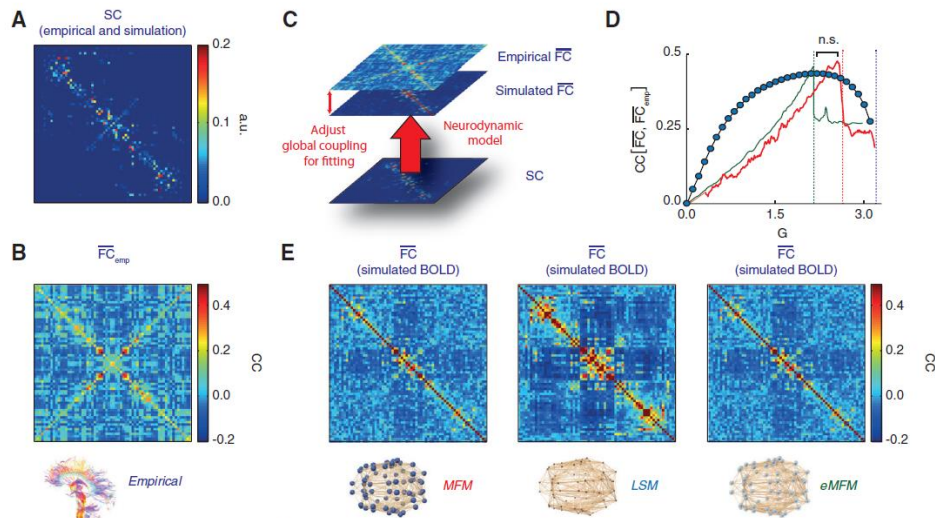
$$\frac{dS_i^I}{dt} = -\frac{S_i^I}{\tau_{NMDA}} + r_i^I + \sigma\eta_i(t)$$

$$r_i^E = \frac{g_E(I_i^E - I_{thr}^E)}{1 - \exp(-d_E g_E(I_i^E - I_{thr}^E))}$$

$$r_i^I = \frac{g_I(I_i^I - I_{thr}^I)}{1 - \exp(-d_I g_I(I_i^I - I_{thr}^I))}$$

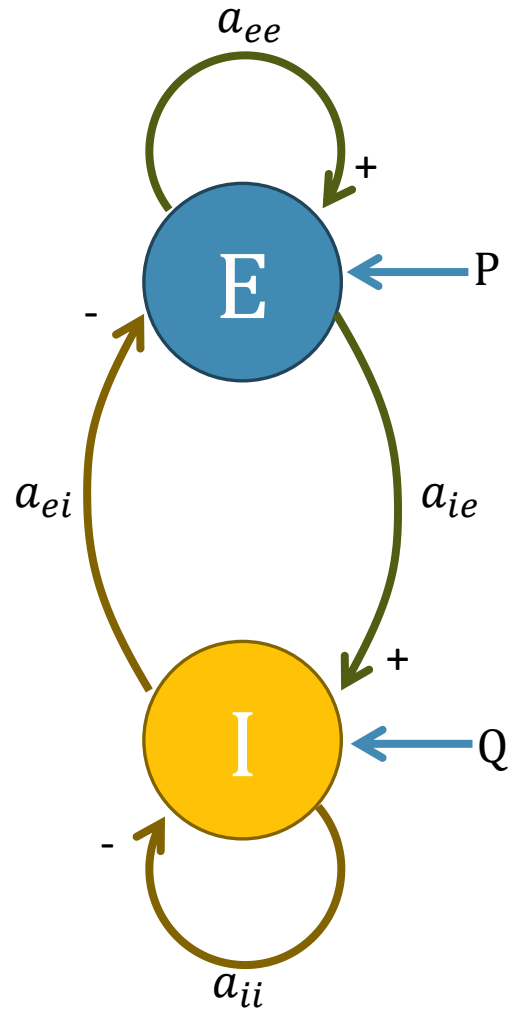
$$I_i^E = W_E I_0 + w_+ J_{NMDA} S_i^E + J_{NNMDA} G \sum_j C_{ij} S_j^E - J_i S_i^I$$

$$I_i^I = W_I I_0 + J_{NMDA} S_i^E - S_i^I$$



Hansen et al., 2015

Wilson-Cowan



- 2 neuron populations: excitatory and inhibitory
- Variables (E,I) represent the firing rate of each population.

$$\tau_e \frac{dE}{dt} = -E + (1 - r_e E) F_e(\alpha_{ee} E - \alpha_{ei} I + P)$$

$$\tau_i \frac{dI}{dt} = -I + (1 - r_i I) F_i(\alpha_{ie} E - \alpha_{ii} I + Q)$$

$$F(x) = \frac{1}{1 + \exp(-\beta(x - x_T))}$$

The role of network structure and time delay in a metapopulation
Wilson–Cowan model

Federica Conti^{a,b}, Robert A. Van Gorder^{a,c,*}



10302–10307 | PNAS | June 23, 2009 | vol. 106 | no. 25

Key role of coupling, delay, and noise in resting brain fluctuations

Gustavo Deco^{a,b,1}, Viktor Jirsa^{c,d}, A. R. McIntosh^e, Olaf Sporns^f, and Rolf Kötter^g

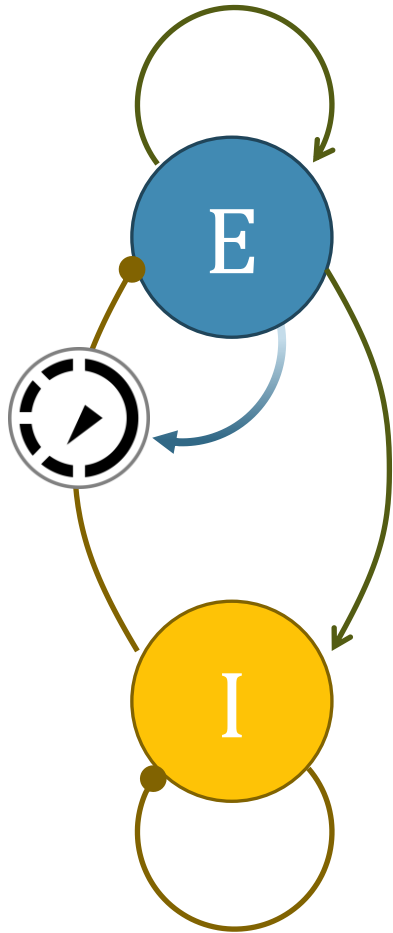
EXCITATORY AND INHIBITORY INTERACTIONS IN LOCALIZED POPULATIONS OF MODEL NEURONS

HUGH R. WILSON *and* JACK D. COWAN

*From the Department of Theoretical Biology, The University of Chicago,
Chicago, Illinois 60637*

BIOPHYSICAL JOURNAL VOLUME 12 1972

Wilson-Cowan w/homeostatic plasticity



- 2 neuron populations: excitatory and inhibitory
- Variables (E,I) represent the firing rate of each population.
- a_{ei} varies dynamically to make E maintain an average value ρ_E
 - $\tau_p \gg \tau_e, \tau_i$

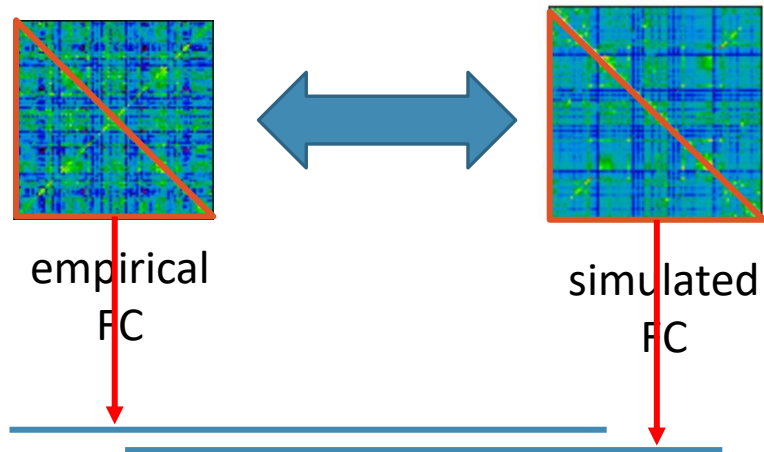
$$\tau_e \frac{dE}{dt} = -E + (1 - r_e E) F_e(a_{ee} E - a_{ei} I + P)$$

$$\tau_i \frac{dI}{dt} = -I + (1 - r_i I) F_i(a_{ie} E - a_{ii} I + Q)$$

$$\tau_p \frac{da_{ei}}{dt} = I(E - \rho_E)$$

$$F(x) = \frac{1}{1 + \exp(-\beta(x - x_T))}$$

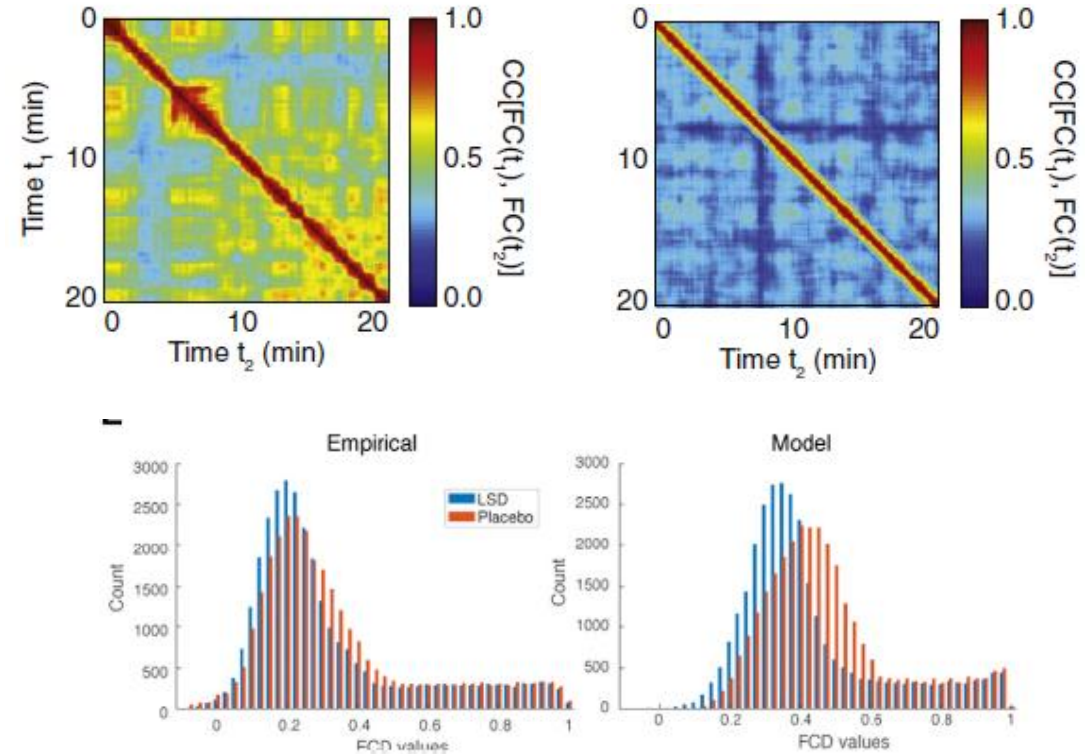
Fitting the model to empirical data



Possible distances to minimize:

- Euclidean Distance
- Correlation (1-corr)
- SSIM (1-ssim)

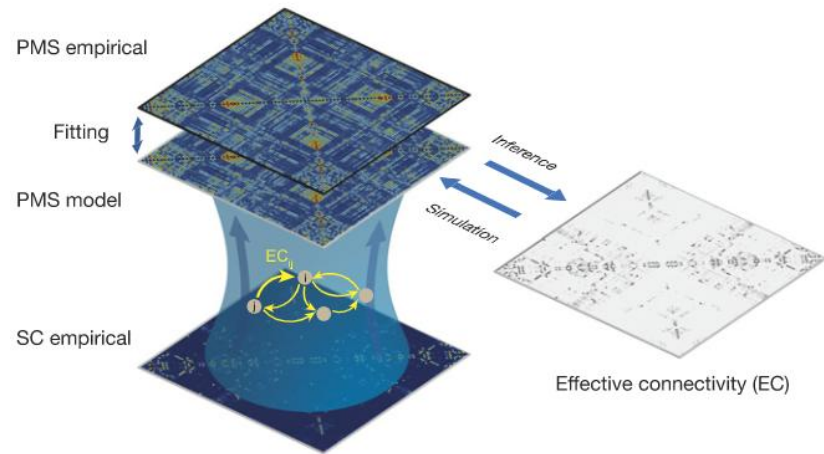
$$\left(\frac{2\mu_x\mu_y+0.01}{\mu_x^2+\mu_y^2+0.01} \right) \left(\frac{2\sigma_x\sigma_y+0.03}{\sigma_x^2+\sigma_y^2+0.03} \right) \left(\frac{\sigma_{xy}+0.015}{\sigma_x\sigma_y+0.015} \right)$$



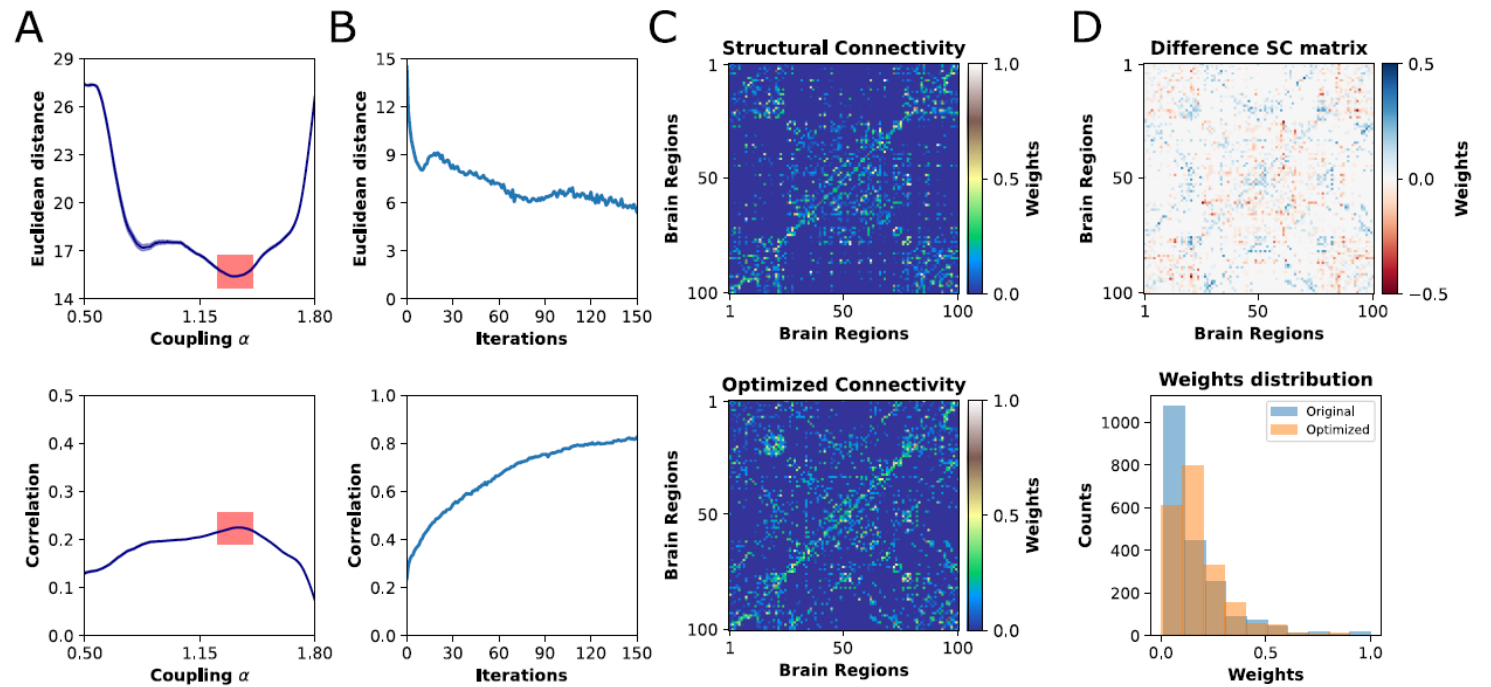
KS distance

Optimizing the SC

- **Problem:** SC and FC do not come from the same individual(s)
- **Solution:** optimize the SC according to differences between empirical and simulated FCs



Deco et al., *PNAS* 2019



Coronel-Oliveros et al., *NeuroImage* 2023

Further topics

- Learning / plasticity
 - Dynamically changing a model parameter, following a learning rule.
 - Typically, connectivity parameters. (they become variables, not params)
- Neural Field models
 - Consideration of spatial propagation of activity (voltage and/or firing rate) using PDEs.
- Dynamic causal modeling
 - Bayesian approach to fit the model to a given data of neural activity
- Hybrid continuous/discrete models
 - Modeling abrupt transitions (states) as a Markov process.